Supplementary Approaches for Enhancing LUNA-CiM Scalability

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I. Introduction

LUNA-CiM achieves 73% area, 58% energy, and 36% speed gains via D&C [1]. LUTs grow exponentially with bit-width, limiting precision beyond 8-bit. This paper presents five complementary approaches leveraging sparsity (40-50% zero in ReLU), heterogeneous precision (7-bit suffices for VGG11 on CIFAR-10), and emerging memory ($10\times$ density vs SRAM). Implementation strategies and quantified benefits are provided. These approaches collectively address scalability while maintaining core LUT advantages in speed and noise margin. The D&C strategy is extended through input-awareness, reconfigurability, stochastic approximation, hybrid memory, and architectural optimization.

II. Input-Aware D&C Activation

Uniform activation wastes energy on zero operands which constitute 50% of ReLU outputs. **Sparsity Exploitation:** Zero-skipping bypasses LUT lookups for zero operands. Implementation: 4-input NOR tree detectors at each D&C input with clock gating to disable LUT wordline and MUX select. Output mux selects constant zero instead of LUT data. Reduces energy 15-25%, area: $4 \times (N/4)$ NOR gates. **Entropy Compression:** Frequent results in SRAM; uncommon in hierarchy. Implementation: 2-bit saturating counters per entry with LRU eviction. Hot entries promoted to SRAM; cold to lower memory. Reduces SRAM 40-60% for typical CNN workloads.

III. Reconfigurable Decomposition Depth

Fixed decomposition creates uniform overhead regardless of precision. **Precision-on-Demand:** Adapt D&C levels via 2-bit config registers (depth 1-4) with bypass MUXes. Depth-1 for 4-bit; depth-2 for 8-bit; depth-4 for 16-bit. Configuration loaded during deployment. Reduces area 40% for mixed-precision. **Layer Optimization:** Profile sensitivity during training using quantization-aware frameworks. VGG11 shows 7-bit inputs suffice. Layers with $<1\%$ degradation receive lower precision, reducing D&C depth. This achieves excellent accuracy-hardware trade-off.

IV. Stochastic Recombination Logic

Approximate variants enable additional savings via controlled statistical errors. **Probabilistic Adders:** Replace deterministic adders with 6T comparator + 4-bit LFSR supporting 0%,25%,50%,100% rounding. LFSR generates random bit; comparator decides round-up/down. Statistically unbiased over many ops. Reduces recombination area 30-40%. **Accuracy-Aware:** Running error accumulator with layer-specific thresholds. When cumulative error exceeds threshold, probability reduces to correct bias. Aggressive (50-100%) for high-tolerance layers; full precision for sensitive layers.

V. Hybrid Memory Hierarchy Integration

NVM overcomes SRAM density limits for larger LUTs. **RRAM/STT-MRAM:** 4KB SRAM L1 cache + 64KB RRAM L2 per unit with TLB for address mapping. LRU promotion from RRAM to SRAM; TLB miss fetches from RRAM and promotes. Maintains SRAM speed with RRAM density. **Capacity:** 8-bit SRAM index (256 entries) + 12-bit RRAM index (4096 entries): $16\times$ capacity, 5% area for cache logic. Partitions LUT space into hot (SRAM) and cold (RRAM) regions. Enables practical 16-bit operations.

VI. D&C Variant Optimization

Alternative D&C variants yield architectural benefits. **Non-Uniform Partitioning:** Asymmetric splitting (e.g., 6-bit $\times$ 4-bit). Implementation: programmable 3-bit ratio (2:2,3:1,4:0,1:3) with configurable shifters. Reduces MSB LUT complexity. **Shared Recombination:** Round-robin scheduler (12-state FSM) sharing one adder across 4 D&C blocks: $4\times$ adder area reduction with 4-cycle latency. Each block holds partial results in local registers until adder available. Acceptable in pipelined accelerators.

VII. Conclusion

These five complementary directions—input-aware activation, reconfigurable decomposition, stochastic recombination, hybrid memory, and D&C variants—collectively address LUNA-CiM's scalability challenges. Quantitative analysis suggests 20-30% additional area-energy-delay improvement with 0-5% overhead. Approaches are combinable for multiplicative benefits. Future work includes 28nm test-chip validation, integration with TensorFlow/PyTorch training frameworks, and exploration of FFT and cryptographic applications where LUT-based computation offers similar advantages in area and energy efficiency.

References

[1] P. Dehghanzadeh, O. Sen, B. Chatterjee, and S. Bhunia, "LUNA-CiM: A programmable compute-in-memory fabric for neural network acceleration," *IEEE Trans. Comput.*, vol. 74, no. 4, pp. 1348–1361, Apr. 2025.